

EDUCATION

Anand Niketan School	Ahmedabad, India
Indian School Certificate (ISC) Grades 11 (82%) and 12 – results pending	
Subjects: Computer Science, Physics, Chemistry, Mathematics, English Language, English Literature, and Mass Media Communication	
June 2025 – May 2027	
Indian Certificate of Secondary Education (ICSE) Grades 9(85%) and 10 (96%)	
Subjects: Computer Applications, Mathematics, Physics, Chemistry, Biology, English Literature, English Language, History and Civics, Geography, and Hindi	
June 2023 – May 2025	

AWARDS AND HONORS

• Crest Gold Award for the project Integrated Assistive System for the Visually Impaired	2026
• Honorable mention: Internationa Mathematical Modelling Challenge - India version 2026	2026
• CISCE Interschool Robotics Competition (IRC) SubNationals (Mumbai)	2024
• Vivo Ignite SubNationals (Top 200 out of 4200 teams)	2023

COMPUTER SCIENCE EXPERIENCES AND PROJECTS

Internship - DRONA Cyber Solutions	Ahmedabad, India
Core Cybersecurity and SOC intern	May 2026 - June 2026
- Completed a cybersecurity internship with hands-on exposure to Security Operations Center (SOC) operations and security workflows. - Built a foundation in network security, Vulnerability Assessment and Penetration Testing (VAPT), and basic ethical hacking techniques. - Explored multiple domains of digital forensics, including cloud, mobile, and memory forensics, and gained practical insight into threat detection and incident response. - Developed an understanding of real-world security challenges and the tools and methodologies used to address them	
Integrated Assistive System for the Visually Impaired – Applied Machine Learning and Artificial Intelligence	
Published - “An Integrated Jacket–Helmet Assistive System for Visually Impaired Individuals Using YOLO-Based Object Detection, Depth Estimation, and OCR” - American Journal of Computer Science and Technology	April – October 2025
- Developed a custom Python pipeline on a Raspberry Pi 4B and implemented YOLOv8 after evaluating various models to deliver 93% accuracy and 15 frames per second for detecting surrounding objects - Achieved under 0.5-second system latency by writing scalable code that integrated depth estimation, optical character recognition, and audio modules into a single edge device - Secured a design patent for a paired jacket and helmet design, which is also low-cost, validating the system with visually impaired individuals to achieve an 84.6 System Usability Scale score during real-world navigation trials [published]	
Low-Cost Brain State Classification - Brain Computer Interfaces and Machine Learning	
Published - “Design and benchmark evaluation of a low-cost single-channel EEG framework for attention–relaxation classification” - Journal of Engineering Research	October - February 2026
- Developed a sub-\$35 wearable electroencephalogram (EEG) system, reducing hardware costs by over 94% versus standard commercial devices while capturing raw cognitive signals at 80-100 Hz - Curated a dataset of 400 seconds from 10 participants, utilizing standard serial communication to log raw analog-to-digital signals directly without applying hardware offset corrections - Evaluated five machine learning models on raw data to achieve 56% test accuracy, demonstrating a measurable 6-point improvement over random chance with 0.55 ROC-AUC values - Demonstrated practical feasibility by integrating the predictive models into an online car simulator, utilizing classified attention and relaxation states to control acceleration and braking dynamically	

PUBLICATIONS

1. K. Ruparelia, P. Parikh, P. A. Shah, and D. Roy, "Design and benchmark evaluation of a low-cost single-channel EEG framework for attention-relaxation classification," *J. Eng. Res.* , in press, doi: 10.1016/j.jer.2026.06.022.
2. K. Ruparelia, P. Parikh, and P. A. Shah, "An integrated jacket-helmet assistive system for visually impaired individuals using YOLO-based object detection, depth estimation, and OCR," *Am. J. Comput. Sci. Technol.* , vol. 8, no. 4, pp. 189–205, 2025, doi: 10.11648/j.ajest.20250804.13.

PATENT

Integrated Assistive System for the Visually Impaired – Applied Machine Learning and AI

- Recognized for innovation with a **granted patent from Intellectual Property India** for the system's design and methodology.

LEADERSHIP AND EDUCATION

Hyperbloom Hacks [\[website\]](#)

Founder

September 2025 – Present

- Founded a virtual hackathon initiative to address gaps in youth computer science education, scaling reach to 220 students across 19 countries through structured online and offline events
- Led a team of four to coordinate and conduct the events, securing 19,000 rupees in funding to incentivize participation and innovation focused on financial literacy and sustainability
- Invited industry professionals, including a senior Amazon engineer, to mentor participants and evaluate submissions, yielding impactful student projects like rural education tools and AI-driven debugging programs.

TIDE Foundation Social Changemaker Fellowship

Junior Fellow

September 2025 – Present

Senior Prefect- Anand Niketan

2022-23

SPORT ACHIEVEMENTS

National Soccer Player

CISCE National Games

2023, 2024, 2025

- secured runners up nationwide in 2024
- Represented the Northwest Region - 3 states, girls u17, right winger, no. of assists, mentoring juniors

Athletics Federation of India Certified Athlete

- Recognised by AFI, one of the highest recognitions in India - track & field and javelin

Khel Mahakumbh (State level Multi-Sport Championship, Organized by Government of Gujarat)

2022

- secured 3rd Position football

ACADEMIC DEVELOPMENT

New York Academy of Sciences – Junior Academy

Student Researcher

September 2025 – December 2025

- Selected for a highly competitive global program (~10% acceptance rate)
- Collaborated with 6 international peers and worked on solar energy infrastructure and storage challenge- my works

Course: IIT Madras CODE - Introduction to Data Science and AI

Course: CodeLQ by IIT Bombay Techfest - Python Programming Beginner

Course: Harvard CS50 Computer Science

Course: Harvard CS50 Cybersecurity

Course: AI Essentials by The Innovation Story

SKILLS & INTERESTS

Technical Skills: Java, Python, Data Science, Computer Vision, Robotics, Embedded Systems, AI/ ML

Interests: Cybersecurity, AI, ML, Brain-computer interface (BCI), LLM and NLP models

Other Skills: Harmonium, Flute, Ukulele, Lawn Tennis, Soccer, Javelin Throw